

Non-visual light modulates behavioral memory and gene expression in *C. elegans*

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Zhijian Ji, Bingying Wang, Rashmi Chandra, Junqiang Liu, Supeng Yang, Yong Long, Michael Egan, Noelle L'Etoile, Dengke K Ma 

Cardiovascular Research Institute, University of California San Francisco, San Francisco, United States • Department of Molecular Cell Biology, University of California, Berkeley, Berkeley, United States • Key Laboratory of Breeding Biotechnology and Sustainable Aquaculture, Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China • Department of Cell and Tissue Biology, University of California San Francisco, San Francisco, United States • Department of Physiology, University of California San Francisco, San Francisco, United States • Innovative Genomics Institute, University of California, Berkeley, Berkeley, United States

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eLife Assessment

This **important** study uncovers a previously unrecognized light-responsive pathway in *C. elegans*, centred on ZIP-2/CEBP-2 and the cytochrome P450 enzyme CYP-14A5. The pathway operates independently of known photoreceptors, modulates long-term memory, and can be harnessed as a low-cost light-inducible expression system, opening new directions for sensory biology and genetic engineering in worms. The strength of evidence is **compelling** if a bacterially derived stimulus is ruled out. Multiple genetic, transcriptional, and behavioural assays support the pathway's role, but a decisive test showing that the initiating light cue is worm-intrinsic rather than mediated by changes in the bacterial food source is still needed.

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Abstract

Visible light influences a range of physiological processes, yet how animals respond to it independently of the visual system remains largely unknown. Here, we uncover a previously undescribed light-induced transcriptional pathway that modulates behavioral plasticity in *C. elegans*, a roundworm without eyes. We demonstrate that ambient visible light or controlled-intensity visible-spectrum LED activates an effector gene *cyp-14A5* in non-neuronal tissues through the bZIP transcription factors ZIP-2 and CEBP-2. Light induction of *cyp-14A5* is more prominent at shorter wavelengths but is independent of the known blue light receptors LITE-1 and GUR-3 in *C. elegans*. This bZIP-dependent genetic pathway in non-neuronal tissues enhances behavioral adaptability and olfactory memory, suggesting a body-brain communication axis. Furthermore, we use the light-responsive *cyp-14A5* promoter to drive ectopic gene expression, causing synthetic light-induced sleep and rapid aging phenotypes in *C. elegans*. These findings advance our understanding of light-responsive mechanisms outside

the visual system and offer a new genetic tool for visible light-inducible gene expression in non-neuronal tissues.

Introduction

Visible light is crucial for image formation and regulating various physiological processes through the visual system, yet how animals respond to ambient light independently of sight remains poorly understood. Recent studies have uncovered diverse non-visual photoreception mechanisms that modulate a range of biological processes, from circadian rhythms to stress responses and metabolic homeostasis (Andrabi et al., 2023 [↗](#); Cronin and Johnsen, 2016 [↗](#); Do and Yau, 2010 [↗](#); Van Gelder, 2008 [↗](#)). These mechanisms often involve specialized light-sensitive proteins, such as opsins and cryptochromes, widely expressed in body locations, including the skin, brain, and peripheral organs. For example, mammalian melanopsin-expressing retinal ganglion cells play critical roles in systemic light responses largely independent of image formation (Berson et al., 2002 [↗](#); Hattar et al., 2003 [↗](#); Lucas et al., 2003 [↗](#); Shi et al., 2024 [↗](#)). In the nematode roundworm *C. elegans*, blue light photoreception requires the light-activated ion channels LITE-1 and GUR-3 in specific neurons, influencing aversive behaviors and cellular physiology (Bhatla and Horvitz, 2015a [↗](#), 2015b [↗](#); Edwards et al., 2008 [↗](#); Gong et al., 2016 [↗](#); Hanson et al., 2023 [↗](#)). Visible light irradiation can also generate photo-oxidative reactive oxygen species in animals (De Magalhaes Filho et al., 2018 [↗](#); Liebel et al., 2012 [↗](#); Mahmoud et al., 2010 [↗](#)). Despite these advances, the molecular pathways and physiological outcomes of non-visual light sensing and responses remain understudied, raising intriguing questions about the mechanistic basis and functional implications of light as an environmental cue beyond vision.

We previously studied how genes encoding cytochrome P450 (CYP) proteins respond to and mediate effects of exposure to environmental stresses in *C. elegans* (Keller et al., 2014 [↗](#); Ma et al., 2013 [↗](#)). Among various transcriptional reporters we generated for CYP-encoding genes to monitor environmental regulation, we serendipitously discovered that the *cyp-14A5* promoter-driven GFP expression is particularly sensitive to ambient visible light exposure. Building upon this initial finding, we conduct transcriptome profiling to identify light-inducible genes in addition to *cyp-14A5*, determine key transcriptional regulators of *cyp-14A5*, and show that the light-inducible CYP-14A5 promotes behavioral plasticity and olfactory memory in *C. elegans*. The findings also provide a genetic tool to use light-inducible *cyp-14A5* promoter to flexibly and ectopically drive gene expression.

Results

Light activates expression of *cyp-14A5* and other genes in *C. elegans*

Cytochrome P450 proteins comprise a highly conserved superfamily of heme-containing monooxygenases critical for metabolizing endogenous and xenobiotic compounds (Denisov et al., 2005 [↗](#)). We constructed transcriptional reporters for genes encoding CYPs in *C. elegans* and found that *cyp-14A5p::GFP* was drastically up-regulated by bright-field transmission light from a microscope inadvertently left on overnight. Using controlled light versus dark conditions, we confirmed the finding from an integrated *cyp-14A5p::GFP* reporter and observed its robust widespread GFP expression in many tissues induced by moderate-intensity (500-3000 Lux, 16-48 hr duration) LED light exposure (Fig. 1A [↗](#)). The level of GFP expression increased proportionally with both light intensity and duration (Fig. 1B [↗](#)), the condition of which does not impact ambient temperature (Fig. S1A [↗](#)-S1D [↗](#)). Other common stresses, including transient 32 °C heat shock,

constant 24-hr hypoxia or starvation did not apparently induce *cyp-14A5p::GFP* (Fig. S1E–S1H). Together, these data suggest that *cyp-14A5p::GFP* induction is finely tuned to ambient light conditions.

To determine CYP-14A5 protein expression pattern, we constructed a translational GFP reporter (*cyp-14A5p::cyp-14A5::GFP*) and observed robust light-induced expression of CYP-14A5::GFP in many of the non-neuronal tissues, including the pharynx, hypoderm and intestine (Fig. 1C). The transcriptional reporter exhibited similar patterns of non-neuronal GFP induction by light (Fig. 1A). The translational reporter reveals the CYP-14A5::GFP pattern indicative of the endoplasmic reticulum structure (Fig. 1C), consistent with the known subcellular localization of most CYPs to ER membranes (Neve and Ingelman-Sundberg, 2010).

To explore how eyeless *C. elegans* responds to ambient visible light independently of a visual system, we conducted transcriptomic profiling by RNA-seq in *C. elegans* exposed to controlled light or dark conditions. We found that defined visible light exposure (1500 Lux, 24 hr duration) to a synchronized population of young adults (24 hrs post L4) triggered a robust genome-wide transcriptional response, including 7902 genes differentially regulated (adjusted *P* value < 0.05, Fig. 1D and Table S1). Among these, *cyp-14A5* was one of the most strongly upregulated genes (Fig. 1E). Gene ontology (GO) analysis of the light-induced transcriptome reveals their significant enrichment in several pathways, including transmembrane signaling, pathogen and stress responses, protein phosphorylation, cellular homeostasis and metabolisms (Fig. S2).

ZIP-2 and CEBP-2 are essential for light-induced transcriptional responses

We next investigated the molecular regulators driving *cyp-14A5* activation in response to light. To test if it requires previously identified blue-light receptors LITE-1 or GUR-3, we crossed the *cyp-14A5p::GFP* reporter with *lite-1* and *gur-3* double loss-of-function mutants. Surprisingly, light-induced GFP expression was preserved in *lite-1gur-3* double mutants, indicating that *cyp-14A5* activation operates through an alternative, non-visual light-sensing mechanism (Fig. 2A). Prolonged photonic light exposure may also cause photo-oxidation of DNA and genotoxicity, leading to DNA damage and ATM protein-dependent check points and transcriptional responses (Ciccio and Elledge, 2010; De Magalhaes Filho et al., 2018; Schuch et al., 2017). However, loss of the DNA damage sensor ATM-1 did not apparently affect light-induced *cyp-14A5p::GFP* (Fig. 2A). These findings underscore the existence of a novel light-responsive pathway in *C. elegans*, distinct from previously characterized photoreceptive systems.

To identify transcriptional regulators driving *cyp-14A5* activation in response to light, we adopted an RNAi-based screening strategy, focusing initially on approximately 400 genes encoding transcription factors, including those responding to various types of stresses. Knockdown of the expression of genes encoding TF from a previously assembled RNAi library or selected for mediating various common stress responses (hypoxia, oxidative stress, heat shock etc.) did not appear to affect *cyp-14A5* activation in response to light (Fig. 2B). Interestingly, we identified from such screen two bZIP-type transcription factors, ZIP-2 and CEBP-2, as critical mediators of light-induced *cyp-14A5* transcription (Fig. 2B). Knockdown or genetic ablation of either *zip-2* or *cebp-2* abolished light-induced *cyp-14A5p::GFP* expression (Fig. 2B, 2C). ZIP-2 and CEBP-2 have been previously identified (Dunbar et al., 2012; Estes et al., 2010; Kniazeva and Ruvkun, 2025; Reddy et al., 2016) to cooperate in a regulatory complex and mediate transcriptional responses to translational inhibition caused by the bacterial pathogen *Pseudomonas aeruginosa* PA14. In these studies, the *irg-1p::GFP* transcriptional reporter was robustly activated by PA14 as a well-established target for ZIP-2 and CEBP-2. Interestingly, we found *irg-1p::GFP* was not activated by the same light condition (1500 Lux, 24 hrs) that reliably induced *cyp-14A5p::GFP* (Fig. 2D). Although ZIP-2 can be activated by pathogen stresses through ribosomal inhibition and subsequent selective ZIP-2 translation (Dunbar et al., 2012; Estes et al., 2010; Kniazeva and

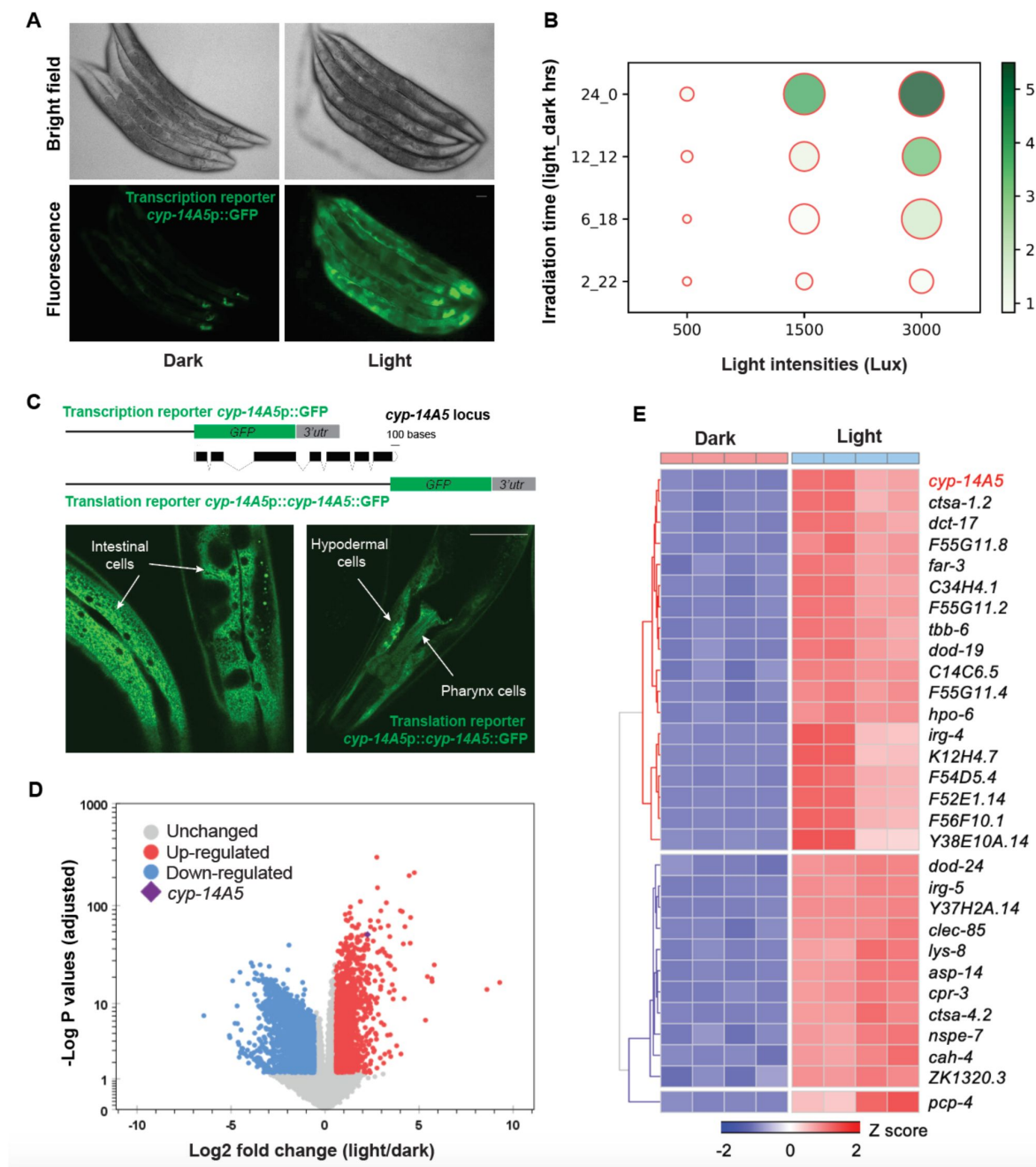


Figure 1

Visible light exposure activates the CYP-encoding gene *cyp-14A5* in a genetic program in *C. elegans*.

A, Representative epifluorescence and brightfield images showing *cyp-14A5p::GFP* induction by visible light exposure (1500 Lux, 24 hrs), in synchronized young adults (24 hrs post L4). Scale bar: 50 μ m. B, Bubble plot showing fold induction of *cyp-14A5p::GFP* as a function of light intensity (Lux) and duration (hours of light_dark indicated in Y axis). C, Schematic of *cyp-14A5* transcriptional and translational GFP reporters. The translational reporter shows non-neuronal (indicated by arrows) induction of CYP-14A5::GFP by light (1500 Lux, 24 hrs). Scale bar: 50 μ m. D, Volcano plot showing genes differentially regulated by light (1500 Lux, 24 hrs), with *cyp-14A5* highlighted, in synchronized young adults (24 hrs post L4). E, Heat map of top-ranking visible light-regulated genes (top 30 including *cyp-14A5*).

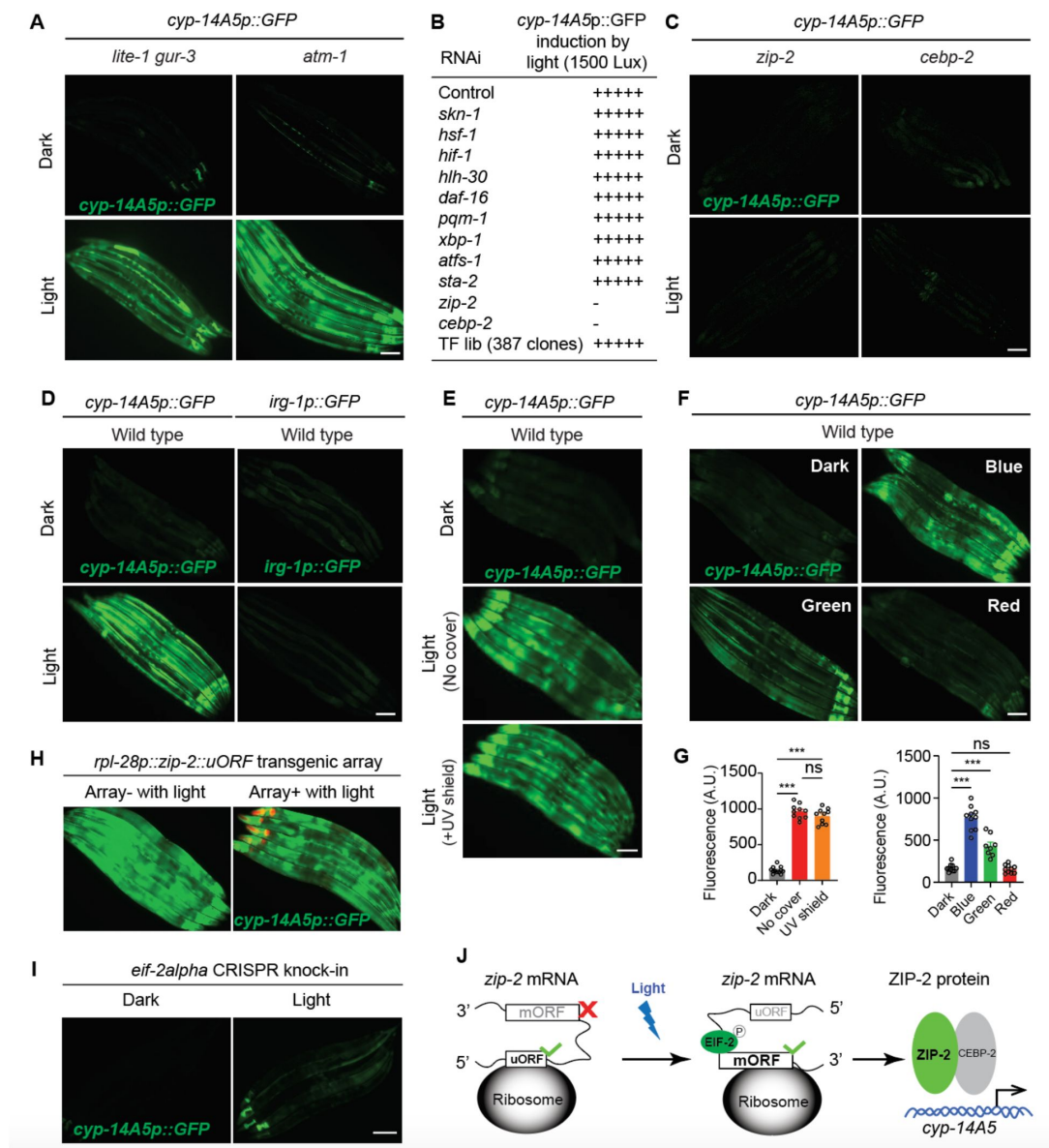


Figure 2

Light induction of *cyp-14A5p::GFP* requires the transcription factors ZIP-2 and CEBP-2.

A, Representative epifluorescence images showing light-induced *cyp-14A5p::GFP* expression in *lite-1 gur-3* double and *atm-1* single loss-of-function mutants. Scale bar: 50 μ m. B, Summary of RNAi screens identifying ZIP-2 and CEBP-2 as essential transcriptional regulators of light-induced *cyp-14A5p::GFP* expression. C, Representative epifluorescence images showing light-induced *cyp-14A5p::GFP* expression in wild type, *zip-2* and *ceb-2* loss-of-function mutants. Scale bar: 50 μ m. D, Representative epifluorescence images showing light-induced *cyp-14A5p::GFP* expression and no light-induced *irg-1p::GFP* expression in wild type animals. Scale bar: 50 μ m. E, Representative epifluorescence images showing light-induced *cyp-14A5p::GFP* expression unaffected by a UV shield. Scale bar: 50 μ m. F, Representative epifluorescence images showing light-induced *cyp-14A5p::GFP* expression by red, green, blue LED light sources of equal intensities. Scale bar: 50 μ m. G, Quantification of E and F. *** indicates $P < 0.001$, n.s., non-significant. H, Representative epifluorescence images showing light-induced *cyp-14A5p::GFP* expression unaffected in *rpl-28p::zip-2uORF* transgenic animals. Scale bar: 50 μ m. I, Representative epifluorescence images showing light-induced *cyp-14A5p::GFP* expression abolished in *eif-2alpha* mutants. Scale bar: 50 μ m. J, Schematic model for light-induced transition of *zip-2* mRNA from translating uORF to mORF, leading to increased ZIP-2 and subsequent increased transcriptional *cyp-14A5* expression in cooperation with CEBP-2. Other molecular players are omitted and uORF and mORF are separated for clarity.

Ruvkun, 2025 [↗](#); Reddy et al., 2016 [↗](#)), our results suggest the specific roles of ZIP-2 in mediating light-induced *cyp-14A5* but not *irg-1* reporter expression, suggesting the involvement of additional stress context-specific factors in these processes.

We further explored conditions and mechanisms leading to light-induced *cyp-14A5p::GFP*. To test potential effects of ultraviolet (UV) irradiation from our visible light LED, we used a UV-masking shield to block UV irradiation. However, this did not affect visible LED light-induced *cyp-14A5p::GFP* expression (**Fig. 2E** [↗](#)). In addition, we found that the LED light exposure of equal intensities (1500 Lux, 24 hrs) but at different wavelengths (red, green, blue) led to differential *cyp-14A5p::GFP* expression (**Fig. 2F** [↗](#), **2G**), showing stronger effects of shorter wavelengths in the visible-light spectrum. A pseudo-open reading frame (uORF) in the 5' untranslated region (UTR) of *zip-2* mRNA inhibits the ribosomal translation of the ZIP-2 main open reading frame (mORF) in the context of PA14 pathogen exposure (Dunbar et al., 2012 [↗](#)). However, constitutive expression of *zip-2* uORF by the *rpl-28* promoter did not affect light-induced *cyp-14A5p::GFP* (**Fig. 2H** [↗](#)). Furthermore, a CRISPR phospho-site knock-in mutation of *eif-2alpha(S49A)* did not affect global translation (Ma et al., 2023 [↗](#)), yet abolished light-induced *cyp-14A5p::GFP* (**Fig. 2I** [↗](#)). As the eukaryotic eIF2alpha complex facilitates translational switch from uORF to mORF upon stress-induced ribosomal stall at uORF (Brito Querido et al., 2024 [↗](#); Costa-Mattioli and Walter, 2020 [↗](#); Mir et al., 2024 [↗](#)), these results suggest that it is the *zip-2* uORF translational inhibition, not the uORF protein product function, that mediates visible light-increased ZIP-2 translation and *cyp-14A5p::GFP* expression (**Fig. 2J** [↗](#)).

Light-induced CYP-14A5 enhances behavioral memory

Although EIF-2alpha and ZIP-2/CEBP-2 functions appear essential for light-induced up-regulation of *cyp-14A5*, the *zip-2* or *cyp-14A5* loss-of-function null mutants show no apparent body-size, morphological, feeding, defecation or developmental defects under dark or LED light treatment (1500 Lux for 16 or 24 hrs) conditions (**Fig. S3** [↗](#)). These data suggest that transient visible light exposure or light-induced *cyp-14A5* activation by ZIP-2 does not broadly impact development, aging or physiology, unlike long-term visible light exposure, which has been shown to robustly shorten lifespans in *C. elegans* (De Magalhaes Filho et al., 2018 [↗](#)).

The lack of obvious morphological and basal behavioral defects led us to explore whether light exposure influences other aspects of *C. elegans* biology, particularly behavioral plasticity and associative memory formation that might require integration of body physiology. Specifically, we chose a learning paradigm in which animals learn to avoid an innately attractive odor butanone after it is paired with aversive stimuli (Chandra et al., 2023 [↗](#); Kauffman et al., 2010 [↗](#)). *C. elegans* can consolidate this learning into a long-lasting memory for up to 16 hours once the repetitive training is followed by sleep and recovery post learning (Chandra et al., 2023 [↗](#)). Using this conditioning protocol (**Fig. 3A** [↗](#)), we observed that animals exposed to ambient light (approximately 500–1000 Lux) during olfactory associative learning and recovery exhibited significantly enhanced memory retention compared to those maintained in darkness (**Fig. 3B** [↗](#)). To pinpoint the critical period for light exposure, we deprived animals of light in two-hour intervals immediately post learning. Interestingly, light deprivation during the first 2–4 hours post learning resulted in markedly impaired memory retention (**Fig. 3C** [↗](#)). These results suggest that environmental light exposure enhances aversive cue association post learning and is not required for learning itself but is required post learning concurrent with sleep for consolidation of memory (Chandra et al., 2023 [↗](#)).

Does the light-modulated behavioral memory consolidation require light activation of the ZIP-2 pathway? To address this question, we examined the behavioral memory of two independent *zip-2* deletion mutants as compared to wild type (*cebp-2* mutants are pleiotropically sick and thus not included). We found that both the *ok3730* and *tm4246* deletion mutations of *zip-2* caused significantly impaired memory consolidation but not learning (**Fig. 3D** [↗](#)). Strikingly, the loss-of-function mutation of *cyp-14A5*, but not *F43C1.7* (another ZIP-2 target gene induced by visible light),

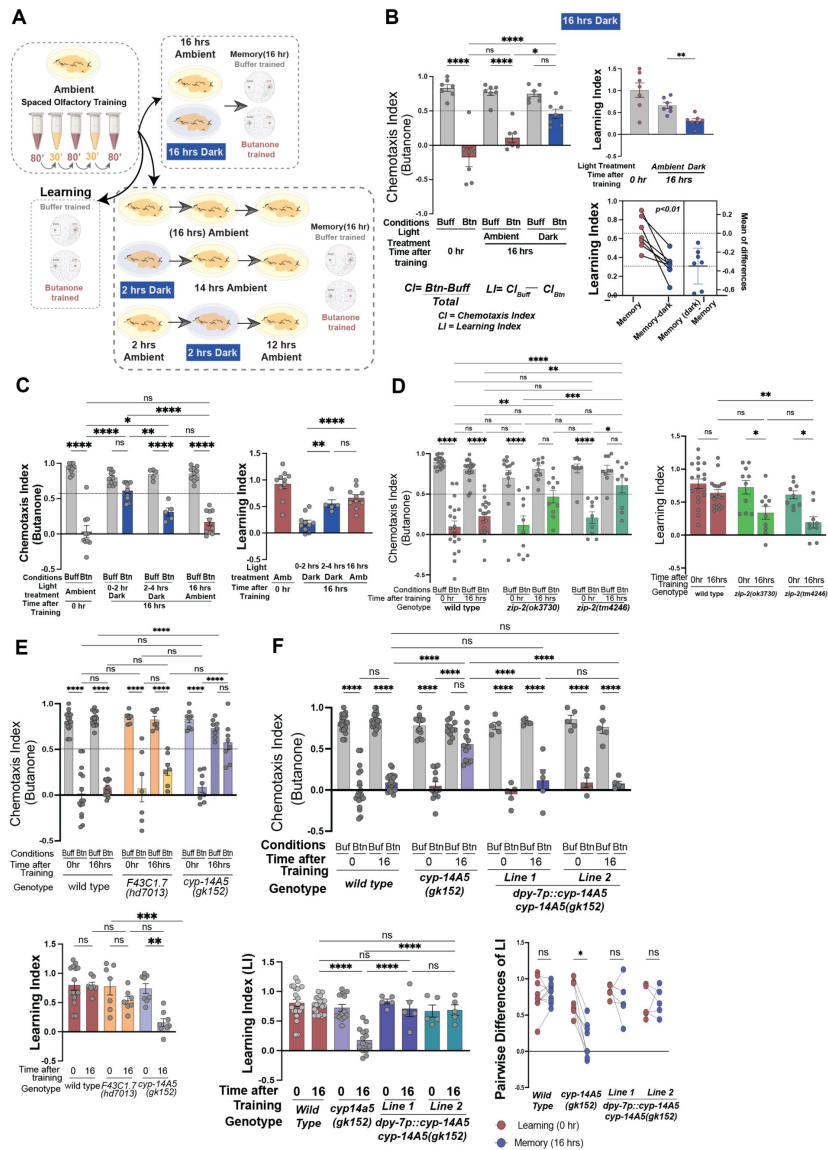


Figure 3

Light promotes behavioral plasticity and memory consolidation via a ZIP-2/CYP axis in hypodermis.

A, Schematic of behavioral setup to test effects of dark/light on olfactory memory. B, Wild-type learning and memory after 16 hours of dark exposure. 7 trials, 50-200 animals per trial/condition. Two-way ANOVA shows significant differences in chemotaxis (CI) under ambient light conditions (approximately 600 Lux) but not when the assay plates were placed in the dark. Learning (LI) indices reflect the differences between buffer- and butanone-treated animals, attenuated under dark (one-way ANOVA). Pairwise t-tests of the amount of memory retention under light and dark recovery reveal the degree of memory loss under dark. C, Dark exposure timeline shows that exposing animals to dark immediately after training (0-2 hr, Dark) hampered memory retention, whereas dark conditions for the 2-4-hour period is less sufficient to induce memory loss (two-way ANOVA). The lack of differences between buffer and butanone-trained animals is reflected in the respective LIs (one-way ANOVA). D, Two different loss-of-function alleles of *zip-2* (*tm4246*) or *zip-2(ok3730)* both showed impaired memory, but learning remained intact (two-way ANOVA). 5-10 trials, 50-200 animals per trial/condition. E, Memory impairment of *cyp-14A5(gk152)* but not *F43C1.7* null mutant animals (two-way ANOVA for CIs and One-way ANOVA for LIs. 7-14 trials, 50-200 animals per trial/condition. F, Memory defects of *cyp-14A5(gk152)* mutants are rescued by hypodermal expression of wild-type *cyp-14A5*. Two independent transgenic lines show similar results with comparison of pairwise differences in LIs and the amount of memory rescued by hypodermal *cyp-14A5* (Cis: Two-way ANOVA; Lis: One-way ANOVA). * indicates $P < 0.05$, ** indicates $P < 0.01$, *** indicates $P < 0.001$, **** indicates $P < 0.0001$, n.s., non-significant.

also impaired memory retention as *zip-2* (Fig. 3E), indicating a crucial role of the ZIP-2/CYP-14A5 regulatory axis in mediating light-modulated memory consolidation. To further delineate the role of CYP-14A5, we performed tissue-specific rescue experiments in the behavioral memory assay. We found that hypoderm-specific expression of *cyp-14A5* restored the behavioral memory in the *cyp-14A5* mutant (Fig. 3F). We observed similar degrees of rescue by two independently derived lines expressing hypoderm-specific *dpy-7p::cyp-14A5* transgenes. These findings strongly suggest that hypodermal induction of CYP-14A5 by ZIP-2 plays a central role in mediating light-modulated behavioral memory.

The *cyp-14A5* promoter as a versatile tool for light-inducible gene expression

The light-responsive nature of the *cyp-14A5* promoter prompted us to explore its potential as a tool for controlling gene expression. We generated synthetic constructs driving the expression of diverse effectors under the *cyp-14A5* promoter to confer striking organismal phenotypes. We previously found that *zip-10* expression promotes organismal phenoptosis (Pandey and Ma, 2022; Wang et al., 2023). Driven by the *cyp-14A5* promoter, light-induced *zip-10* expression indeed caused a robust light-dependent rapid aging or phenoptosis-like phenotype with markedly shortened median and maximal lifespans (Fig. 4A–4D). We confirmed that LED light exposure markedly induced *zip-10* expression, as evidenced by robust ZIP-10-tagging mCherry fluorescence in major non-neuronal tissues of transgenic animals, only after light (1500 Lux, 24 or 48 hrs) exposure (Fig. 4C).

To test organismal behavioral outcomes, we expressed *nlp-22* under the *cyp-14A5* promoter. *nlp-22* was previously identified as a sleep-promoting neuropeptide (Bringmann, 2018; Nelson et al., 2013; Van der Auwera et al., 2020), overexpression of which can cause drastic reduction of pumping and locomotion speed, characteristic of sleep behaviors in *C. elegans*. We found that *cyp-14A5p::nlp-22* can indeed trigger striking behavioral quiescence upon light exposure (1500 Lux, 24 or 48 hrs), as quantified by pumping rates, bending frequencies and locomotion speed (Fig. 4E–4G). Quantitative analysis by WormLab reveals that the behavioral quiescence induced by *nlp-22* corresponded to characteristic bouts of sleep (Fig. 5A). These proof-of-concept studies demonstrate that the *cyp-14A5* promoter enables light-dependent ectopic induction of gene expression, offering a flexible tool for probing gene function, studies of organismal biology, behaviors and synthetic physiology applications.

Discussion

Our findings uncover a previously unknown light-induced transcriptional pathway in *C. elegans* that operates independently of known visual light receptors. Our study also establishes a functional link between ambient light and behavioral plasticity through a ZIP-2/CYP regulatory axis. The discovery of this pathway and its organismal functions opens exciting avenues for understanding body-brain communication and how environmental cues such as light can shape physiological and behavioral processes.

The specificity of this pathway is particularly intriguing, as *cyp-14A5* is robustly induced by light at wavelength and intensity that do not apparently alter ambient temperature (Fig. S1). Previous studies identified *cyp-14A5* as one of many genes moderately regulated by bacterial pathogens (Sinha et al., 2012; Troemel et al., 2006; Wong et al., 2007), yet the classic pathogen-inducible gene reporter *irg-1p::GFP* was not apparently activated by light conditions that induced *cyp-14A5p::GFP*. This specificity raises important questions about how ZIP-2 and *cyp-14A5* are selectively activated by light, or to different degrees. Given the crucial roles of ZIP-2 and eIF2alpha we discovered for light-induced expression of *cyp-14A5* and the established role of the uORF at the 5' untranslated region of *zip-2* RNA in ZIP-2 regulation (Dunbar et al., 2012), it is plausible that

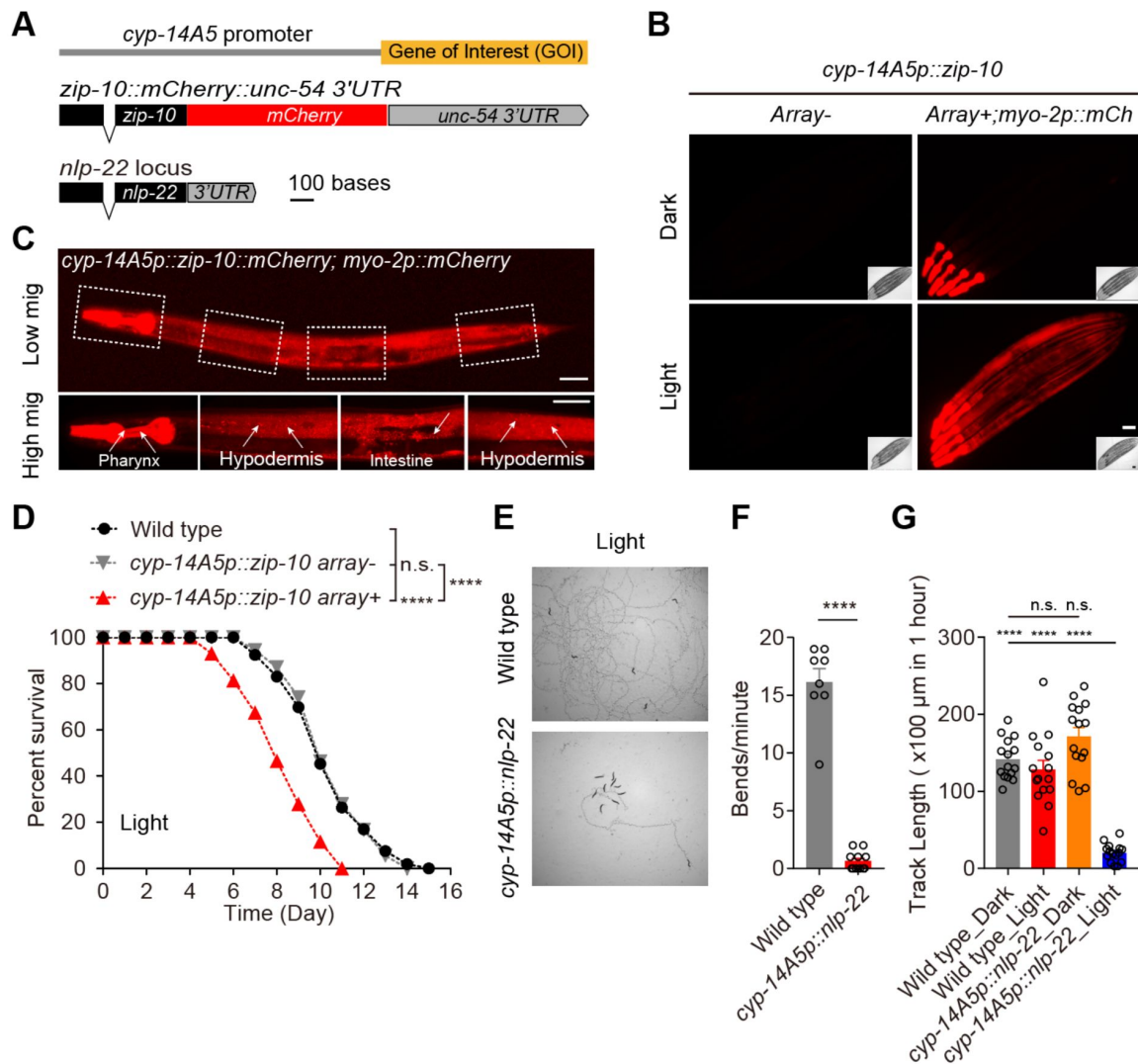


Figure 4

Light-induced gene expression drives organismal phenotypes, including sleep and shortened lifespans.

A, Schematic of synthetic constructs for light-inducible *nlp-22* and *zip-10::mCherry* using the *cyp-14A5* promoter. B, Representative compound epifluorescence images showing light-induced *cyp-14A5p::zip-10::mCherry* activation. Scale bar: 50 μ m. Pharyngeal muscle-specific *myo-2p::mCherry* was used as a co-injection marker. C, Representative confocal fluorescence images showing light-induced *cyp-14A5p::zip-10::mCherry* activation (1500 Lux, 24 hrs starting at 24 hrs post L4) in major non-neuronal tissues (hypoderm, intestinal cells indicated by arrows). Scale bar: 50 μ m. D, Representative lifespan curves showing that light-induced *zip-10* can markedly shorten lifespan through transient light exposure (1500 Lux, 24 hrs starting at 24 hrs post L4). **** indicates $P < 0.0001$. E, Representative bright field images showing quiescent sleep behaviors by light-induced *cyp-14A5p::nlp-22* expression through transient light exposure (1500 Lux, 24 hrs starting at 24 hrs post L4). F, Quantification of population bending frequencies for transient light-treated (1500 Lux, 24 hrs starting at 24 hrs post L4) control wild type and *cyp-14A5p::nlp-22* animals. G, Quantification of population track lengths for control wild type and *cyp-14A5p::nlp-22* animals with transient light (1500 Lux, 24 hrs starting at 24 hrs post L4) or darkness treatments. *** indicates $P < 0.001$, **** indicates $P < 0.0001$, n.s., non-significant.

light may regulate ZIP-2 translation by *zip-2* RNA photo-oxidation at specific sites, eIF2alpha phosphorylation and specialized ribosomal signaling (D’Orazio and Green, 2021 [DOI](#); Genuth and Barna, 2018 [DOI](#); Sinha et al., 2024 [DOI](#)). Further investigation into the upstream signaling events and the molecular sensors linking light exposure to ZIP-2/CEBP-2 activation is warranted.

Our behavioral assays demonstrate that light exposure enhances olfactory associative memory, providing direct evidence for the functional relevance of light-induced *cyp-14A5* expression. Interestingly, light exposure appears to exert its effects during a specific temporal window hours following learning, suggesting that light-induced transcriptional changes post learning play a key role in memory consolidation. The discovery of *cyp-14A5* as a key effector in this pathway also provides new insights into how non-neuronal tissues contribute to behavioral plasticity. Our findings suggest that light-induced CYP-14A5 and CYP-dependent metabolic or signaling changes in the hypoderm may communicate with the nervous system to influence behavioral memory. This body-brain communication axis highlights the importance of systemic integration in mediating complex physiological and behavioral responses to environmental cues (Aghayeva et al., 2021 [DOI](#); Fukuda et al., 2025 [DOI](#); Liu et al., 2022 [DOI](#); Zhang et al., 2018 [DOI](#)).

Beyond its biological significance, the light-inducible *cyp-14A5* promoter offers a useful new tool for gene expression studies in *C. elegans*. The ability to drive ectopic gene expression in response to light provides a versatile system for temporally controlled genetic manipulations. Our demonstration of light-induced sleep and mortality phenotypes through ectopic gene expression illustrates the potential applications of this tool in studying diverse biological processes in synthetic biology and physiology.

Previous studies have used heat shock or drug-inducible promoters for temporally controlled gene expression in *C. elegans* (Monsalve et al., 2019 [DOI](#); Stringham et al., 1992 [DOI](#); Wei et al., 2012 [DOI](#)). The light-inducible *cyp-14A5* promoter provides an alternative, simple-to-implement approach and might be particularly useful when the drug-inducible system is cumbersome, or heat shock effects are undesirable.

While our study uncovers a novel light-responding mechanism with functional consequences in *C. elegans*, several limitations exist. First, the precise molecular mechanism by which visible light activates ZIP-2 and/or CEBP-2 remains unclear, as does the upstream signaling cascade linking light exposure to transcriptional activation. Second, although we demonstrate a functional connection between light-induced *cyp-14A5* expression and behavioral outcomes, the exact molecular interplay between peripheral transcriptional changes and neural plasticity requires further exploration. Finally, while the *cyp-14A5* promoter serves as a useful genetic tool, it does not confer tissue specificity, and its ectopic effector expression requires control for light effects. These limitations provide fertile ground for future research to build upon our findings.

Materials & methods

C. elegans strains

C. elegans strains were grown on nematode growth media (NGM) plates seeded with *Escherichia coli* OP50 at 20 °C with laboratory standard procedures unless otherwise specified. The N2 Bristol strain was used as the reference wild type (Brenner, 1974 [DOI](#)). Mutants and integrated transgenes were back-crossed at least five times. Genotypes of strains used are as follows: *dmaIs156 IV [cyp-14A5p:: cyp-14A5::GFP; unc-54p::mCherry]*, *agIs17 IV [irg-1p::gfp]*, *dmaEx [dpy-7p::cyp-14A5; myo-2p::mCherry]*, *dmaEx [cyp-14A5p::zip-10::mCherry; myo-2p::mCherry]*, *dmaEx [cyp-14A5p::nlp-22; myo-2p::mCherry]*, *ceb-2 (tm5421) I*, *eif-2alpha(rog3) I*, *zip-2(tm4248) III*, *zip-2(ok3730) III*, *cyp-14A5(gk152) V*.

PCR fusion constructs were used to generate transgenes (Hobert, 2002 [\[4\]](#)), using primer sequences:

DM1310_ *cyp-14A5*Pro c5p TCAACCACATCTTCCGATCA;

DM1311_ *cyp-14A5*Pro to GFP c3p

CGACCTGCAGGCATGCAAGCTgatctttgttgacagaatagtttt;

DM2857_ *dpy-7*p to *cyp-14A5* codutr Forward TGTCTCTGACGCCTGTGAGT;

DM2858_ *dpy-7*p to *cyp-14A5* codutr Reverse

GATAAAGCAACGATGAAAACGCTCATTTTGTTTTCACAGAGCGGTAGA;

DM2944_ *zip-2*uORF to *rpl-28*p-GOI-mCherry-5utr fusion F

CATCATAAAATAATTTATTTCCAGGTAAAATGTATCACGCAAAGACAACCACCG;

DM2945_ *zip-2*uORF to *rpl-28*p-GOI-mCherry-5utr fusion

catgttatcttctcaccctttgaggagccAAGCTCCCGTGGGAAGCTTGTG;

DM2948_ *zip-10* to *cyp-14A5*p-GOI-mCherry-5utr fusion F

aaaactattctgtccaacaaagatcaaaATGACAACAATGACTAATTCTCTTATTTTC;

DM2949_ *zip-10* to *cyp-14A5*p-GOI-mCherry-5utr fusion R

catgttatcttctcaccctttgaggagccGGAATGGTTGATTTGATTATTGAGTTG

DM2952_ *nlp-22*cod::3utr to *cyp-14A5*p-GOI fusion

aaaactattctgtccaacaaagatcaaaATGCGTTCCATAATCGTCTTCATCG;

DM2953_ *nlp-22*cod::3utr to *cyp-14A5*p-GOI fusion R cggttccactttctcatgagt

Fluorescence microscopy and imaging

SPE confocal (Leica) and epifluorescence microscopes were used to capture fluorescence images. Animals were randomly picked at the same stage and treated with 1 mM levamisole in M9 solution (31742-250MG, Sigma-Aldrich), aligned on a 2% agar pad on a slide for imaging. Identical setting and conditions were used to compare experimental groups with control. For quantification of GFP fluorescence, animals were outlined and quantified by measuring gray values using the ImageJ software. The data were plotted and analyzed by using GraphPad Prism10.

For light-induced reporter imaging, reporter animals (synchronized young adults, 24 hrs post L4) were exposed to white light (1500 Lux for 16 or 24 hrs, by Viribright 12-Watt, 800 Lumen, LED Desk Lamp Dimmable Office Lamp). For blue, green (SPE confocal, Leica), and red light (HQRP 660 nm 14w LED pure red) conditions, animals of the same stage were exposed to the same intensities (1500 Lux, 16 or 24 hours). Control groups from the same batch of animals were maintained in darkness by opaque shields. Light intensities and temperature were quantitatively measured by digital light meters and thermometers.

RNA sequencing

A synchronized population of wild-type young adult (24 hrs post L4) animals were exposed to LED light (1500 lux, 24 hr duration, Viribright 12-Watt, 800 Lumen, LED Desk Lamp Dimmable Office Lamp). Control groups from the same batch of animals were maintained in darkness by opaque light shields. Four independent biological replicates were used for both light-treated and control groups. For sample collection, the animals were washed down from NGM plates using M9 solution and bacteria-cleaned with M9 washing in centrifuge tubes, homogenized by tissue disruptors and subjected to RNA extraction using the RNeasy Mini Kit from Qiagen. 1 mg total RNA from each sample was used for sequencing library construction. The libraries were constructed and sequenced for paired end 150 bp by DNBseq (Innomics). The cleaned RNAseq reads were mapped to the genome sequence of *C. elegans* using hisat2 and the mapped reads were assigned to the genes using featureCounts (Kim et al., 2015; Liao et al., 2014). The abundance of genes was expressed as RPKM (Reads per kilobase per million mapped reads) and identification of differentially expressed genes (Table S1) was performed using the DESeq2 package (Love et al., 2014).

C. elegans behavioral assays

The olfactory behavioral memory assay was as described previously with modification (Chandra et al., 2023; Kauffman et al., 2010). Briefly, one day old adult worms (24 hrs post L4) were washed with S basal buffer (0.1 M NaCl, 0.05 M K₃PO₄, pH 6.0) off 10 cm NGM plates and into microfuge tubes, where they were washed three times with S basal buffer. The animals were split in two groups; one group was added to a microfuge tube of S basal media and the other group was added to a microfuge tube of 1:10,000 dilution of butanone in S basal. The microfuge tubes were then rotated for 80 minutes. The odor training includes three 80-minute cycles of training with odor, or a control buffer interspersed with two 30-minute periods of feeding with OP50 *E. coli* bacteria. For chemotaxis assay, 1 µL of (1 M) NaN₃ was pipetted onto the odor and diluent spots in 10 cm plastic petri dishes. 1 µL of 200 proof ethanol was added to the diluent spot and 1 µL of 1:1000 butanone was added to the odor spot, while S basal or butanone-trained worms were dropped onto the middle of 10 cm plastic petri dishes. The recovery period was either under darkness or ambient light (approximately 600 Lux) for 16 hrs or were kept under darkness for 2 or 4 hr periods followed by light exposure and chemotaxis to assay behavioral memory.

For sleep analysis induced by *cyp-14A5p::nlp-22*, the bending angles, moving average speed, and track length of *C. elegans* after 48 hours of exposure to either light or dark conditions were analyzed using WormLab. In such experiment, a synchronized population of young adult (24 hrs post L4) animals of indicated genotype or transgene expression were used. After light exposure or control dark treatment, they were transferred to a fresh NGM plate seeded with a small OP50 bacterial lawn and allowed to settle for at least ten minutes to recover at room temperature. After the recovery period, a one-hour recording session was conducted using WormLab. Bending angles were calculated as described in the referenced method as a metric for sleep behaviors (Chandra et al., 2023). Moving average speed was determined by tracking the displacement of the worms over time.

Statistics

Numerical data were analyzed using GraphPad Prism 10 Software (Graphpad, San Diego, CA) and presented as means ± S.D. unless otherwise specified, with *P* values calculated by unpaired two-tailed t-tests (comparisons between two groups), one-way ANOVA (comparisons across more than two groups) and two-way ANOVA (interaction between genotype and treatment), with post-hoc Tukey and Bonferroni's corrections. The lifespan assay was plotted and quantified using Kaplan–Meier lifespan analysis, and *P* values were calculated using the log-rank test.

Supplementary figures

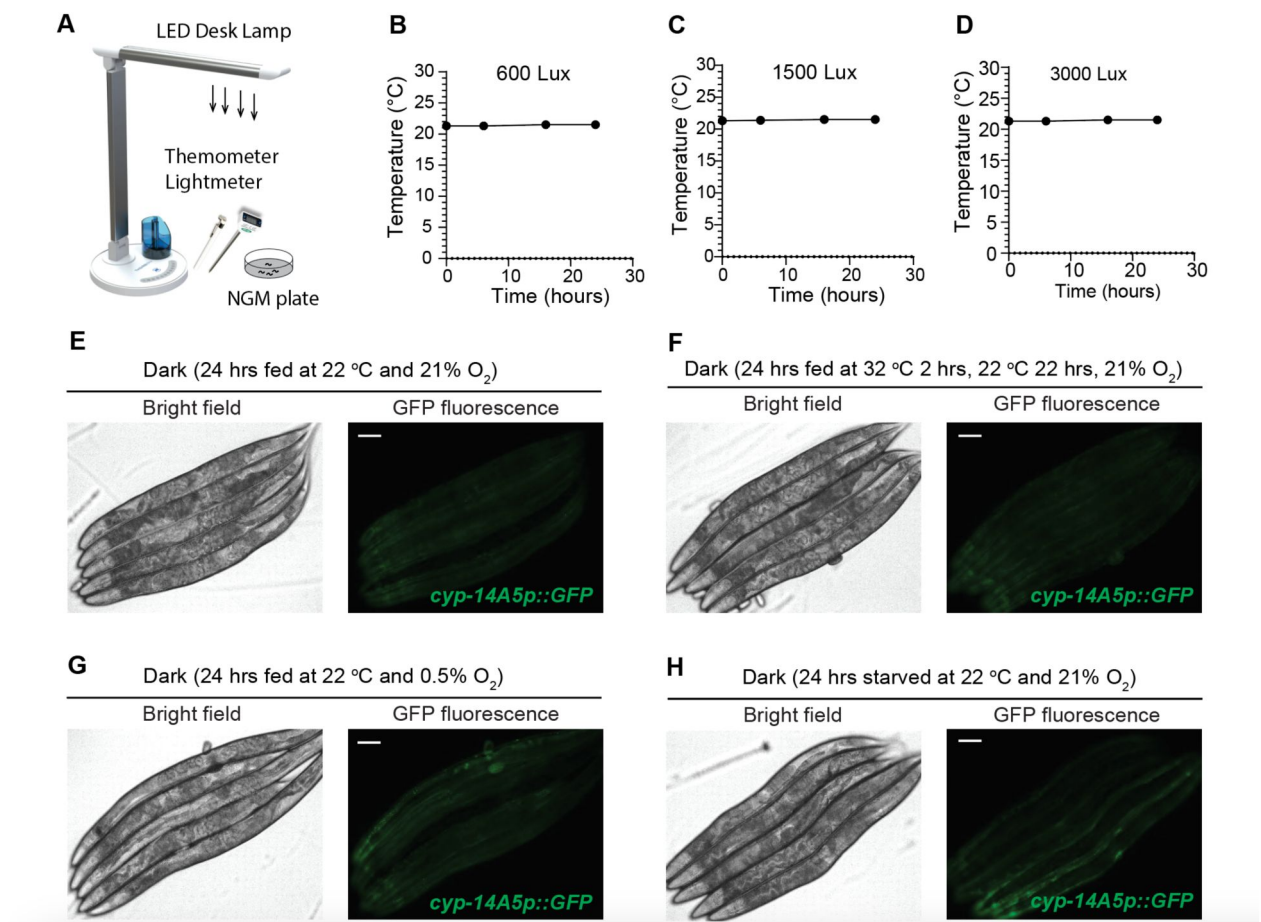


Figure S1

***cyp-14A5p::GFP* induction responds primarily to visible light exposure rather than changes in ambient oxygen, nutrient or temperature.**

A, Schematic of the setup for measurements of light intensity and temperature at a plane (in parallel to LED visible light sources) where animals are exposed to LED light in NGM plates. B, Measurements of temperature at the 600 Lux light intensity showing no change of temperature over 24 hrs. C, Measurements of temperature at the 1500 Lux light intensity showing no change of temperature over 24 hrs. D, Measurements of temperature at the 3000 Lux light intensity showing no change of temperature over 24 hrs. E-H, Representative bright-field and epifluorescence images showing no apparent *cyp-14A5p::GFP* induction by transient exposure to heat shock (F), constant exposure to 24 hrs of hypoxia (G) or starvation (H). Scale bar: 50 μ m.

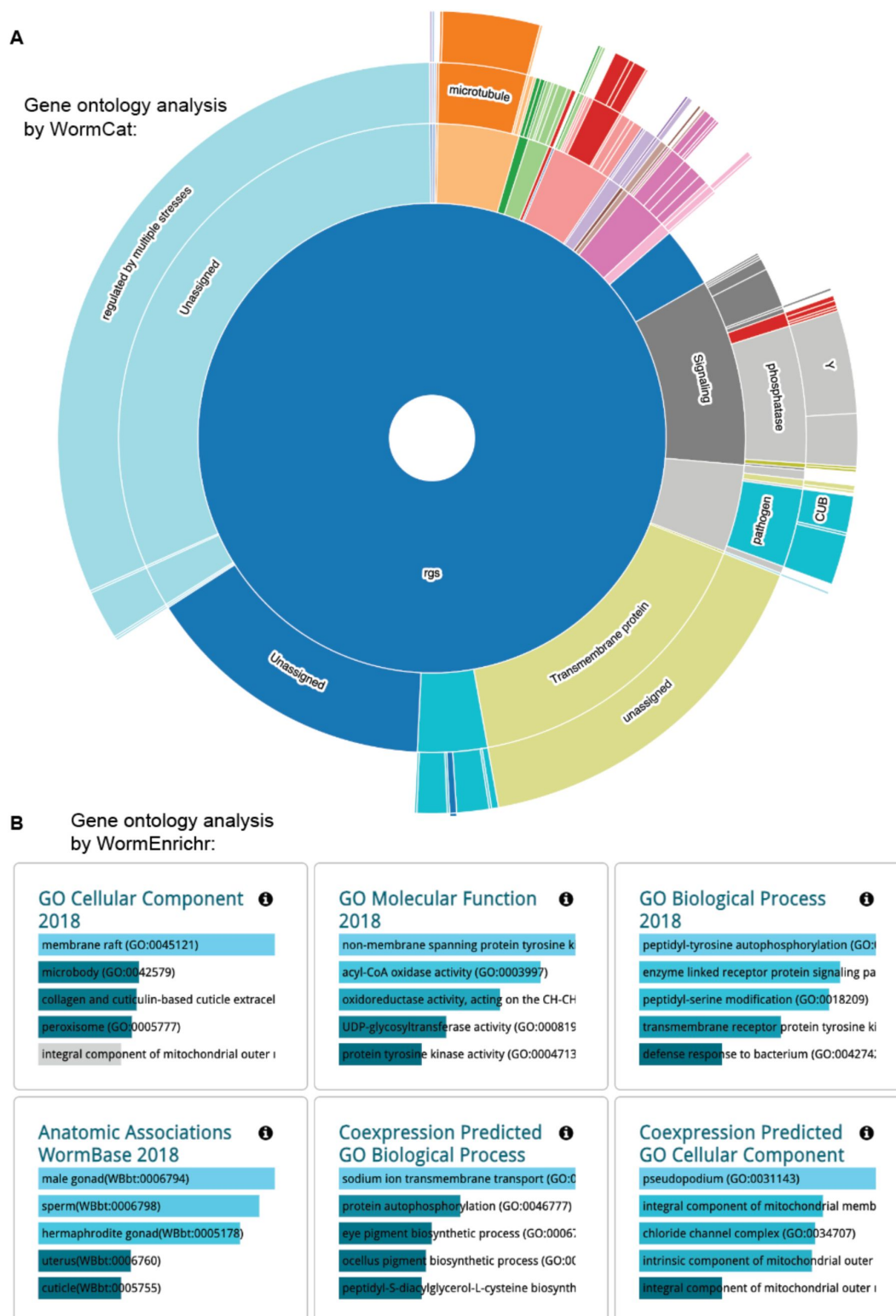


Figure S2

Gene ontology analysis of light-induced transcriptomic changes.

A, Starburst plot of gene ontology analysis of light-induced genes using WormCat (<http://www.wormcat.com/>). B, Table summary of gene ontology analysis of light-induced genes using WormEnrichr (<https://maayanlab.cloud/WormEnrichr/>).

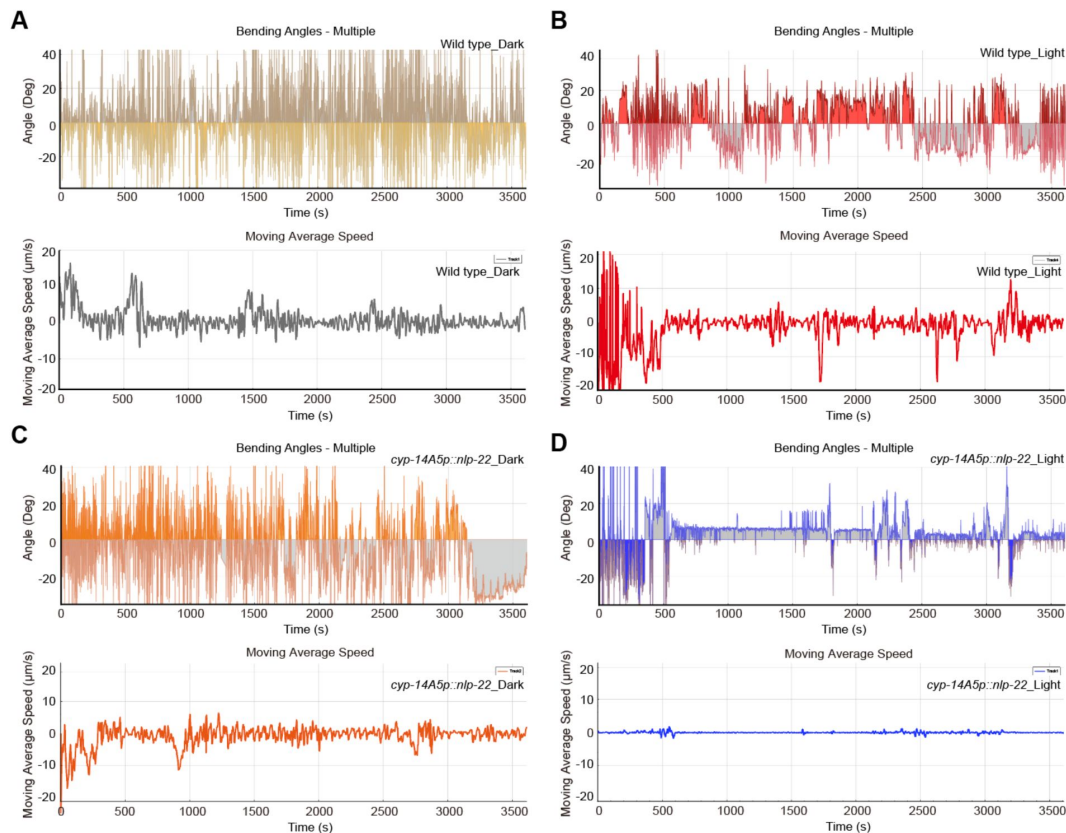


Figure S4

WormLab analysis reveals sleep bouts caused by light-induced *cyp-14A5p::nlp-22* expression.

(A) Representative tracking of moving angles for body posture and average speed for control animals under dark conditions (N=15). (B) Representative tracking of moving angles for body posture and average speed for control animals after light exposure conditions (1500 Lux, 24 hrs, N=13). (C) Representative tracking of moving angles for body posture and average speed for *cyp-14A5p::nlp-22* transgenic animals under dark conditions (N=12). (D) Representative tracking of moving angles for body posture and average speed for *cyp-14A5p::nlp-22* transgenic animals after light exposure conditions (1500 Lux, 24 hrs, N=13).

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Additional information

Author Contributions

D.K.M. designed and analyzed the *C. elegans* experiments, contributed to project conceptualization and wrote the manuscript. B.W. made the initial observation on *cyp-14A5* induction by light and performed RNAi screens and reporter imaging. Z.J. prepared RNA-seq samples and characterized light effects on the *cyp-14A5* reporter, cellular, behavioral and physiological phenotypes of various mutants with assistance from W.Y. and M.E. Y.L. analyzed the RNA-seq data. R.C. designed, performed and analyzed the behavioral memory assays. J.L. designed, performed and analyzed the light-induced sleep and phenotypic aging assays. N.L., Y.L., B.W., Z.J., R.C., J.L. contributed to research materials, project conceptualization and editing manuscript.

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Additional files

Supplementary Table S1 [\(link\)](#)

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Author information

Zhijian Ji[§]

Cardiovascular Research Institute, University of California San Francisco, San Francisco, United States

[§]Equal contribution

Bingying Wang[§]

Cardiovascular Research Institute, University of California San Francisco, San Francisco, United States

[§]Equal contribution

Rashmi Chandra[§]

Cardiovascular Research Institute, University of California San Francisco, San Francisco, United States

[§]Equal contribution

Junqiang Liu[§]

Cardiovascular Research Institute, University of California San Francisco, San Francisco, United States

[§]Equal contribution

Supeng Yang

Cardiovascular Research Institute, University of California San Francisco, San Francisco, United States, Department of Molecular Cell Biology, University of California, Berkeley, Berkeley, United States

Yong Long

Key Laboratory of Breeding Biotechnology and Sustainable Aquaculture, Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China

Michael Egan

Cardiovascular Research Institute, University of California San Francisco, San Francisco, United States, Department of Molecular Cell Biology, University of California, Berkeley, Berkeley, United States

Noelle L'Etoile

Department of Cell and Tissue Biology, University of California San Francisco, San Francisco, United States

Dengke K Ma

Cardiovascular Research Institute, University of California San Francisco, San Francisco, United States, Department of Physiology, University of California San Francisco, San Francisco, United States

Francisco, United States, Innovative Genomics Institute, University of California, Berkeley, Berkeley, United States
ORCID iD: [0000-0002-5619-7485](https://orcid.org/0000-0002-5619-7485)

For correspondence: Dengke.Ma@ucsf.edu

Editors

Reviewing Editor

Sonia Sen

Tata Institute for Genetics and Society, Bangalore, India

Senior Editor

Sonia Sen

Tata Institute for Genetics and Society, Bangalore, India

Reviewer #1 (Public review):

Summary:

The authors set out to understand how animals respond to visible light in an animal without eyes. To do so, they used the *C. elegans* model, which lacks eyes, but nonetheless exhibits robust responses to visible light at several wavelengths. Here, the authors report a promoter that is activated by visible light and independent of known pathways of light responses.

Strengths:

The authors convincingly demonstrate that visible light activates the expression of the *cyp-14A5* promoter-driven gene expression in a variety of contexts and report the finding that this pathway is activated via the ZIP-2 transcriptionally regulated signaling pathway.

Weaknesses:

Because the ZIP-2 pathway has been reported to be activated predominantly by changes in the bacterial food source of *C. elegans* – or exposure of animals to pathogens – it remains unclear if visible light activates a pathway in *C. elegans* (animals) or if visible light potentially is sensed by the bacteria on the plate, which also lack eyes. Specifically, it is possible that the plates are seeded with excess *E. coli*, that *E. coli* is altered by light in some way, and in this context, alters its behavior in such a way that activates a known bacterially responsive pathway in the animals. This weakness would not affect the ability to use this novel discovery as a tool, which would still be useful to the field, but it does leave some questions about the applicability to the original question of how animals sense light in the absence of eyes.

<https://doi.org/10.7554/eLife.108507.1.sa2>

Reviewer #2 (Public review):

Summary:

Ji, Ma, and colleagues report the discovery of a mechanism in *C. elegans* that mediates transcriptional responses to low-intensity light stimuli. They find that light-induced transcription requires a pair of bZIP transcription factors and induces expression of a cytochrome P450 effector. This unexpected light-sensing mechanism is required for physiologically relevant gene expression that controls behavioral plasticity. The authors further show that this mechanism can be co-opted to create light-inducible transgenes.

Strengths:

The authors rigorously demonstrate that ambient light stimuli regulate gene expression via a mechanism that requires the bZIP factors ZIP-2 and CEBP-2. Transcriptional responses to light stimuli are measured using transgenes and using measurements of endogenous transcripts. The study shows proper genetic controls for these effects. The study shows that this light-response does not require known photoreceptors, is tuned to specific wavelengths, and is highly unlikely to be an artifact of temperature-sensing. The study further shows that the function of ZIP-2 and CEBP-2 in light-sensing can be distinguished from their previously reported role in mediating transcriptional responses to pathogenic bacteria. The study includes experiments that demonstrate that regulatory motifs from a known light-response gene can be used to confer light-regulated gene expression, demonstrating sufficiency and suggesting an application of these discoveries in engineering inducible transgenes. Finally, the study shows that ambient light and the transcription factors that transduce it into gene expression changes are required to stabilize a learned olfactory behavior, suggesting a physiological function for this mechanism.

Weaknesses:

The study implies but does not show that the effects of ambient light on stabilizing a learned olfactory behavior are through the described pathway. To show this clearly, the authors should determine whether ambient light has any effect on mutants lacking CYP-14A5, ZIP-2, or CEBP-2. Other minor edits to the text and figures are suggested.

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Reviewer #3 (Public review):

Ji et al. report a novel and interesting light-induced transcriptional response pathway in the eyeless roundworm *Caenorhabditis elegans* that involves a cytochrome P450 family protein (CYP-14A5) and functions independently from previously established photosensory mechanisms. Although the exact mechanisms underlying photoactivation of this pathway remain unclear, light-dependent induction of CYP-14A5 requires bZIP transcription factors ZIP-2 and CEBP-2 that have been previously implicated in worm responses to pathogens. The authors then suggest that light-induced CYP-14A5 activity in the *C. elegans* hypoderm can unexpectedly and cell-non-autonomously contribute to retention of an olfactory memory. Finally, the authors demonstrate the potential for this pathway to enable robust light-induced control of gene expression and behavior, albeit with some restrictions. Overall, the evidence supporting the claims of the authors is convincing, and the authors' work suggests numerous interesting lines of future inquiry.

(1) The authors determine that light, but not several other stressors tested (temperature, hypoxia, and food deprivation), can induce transcription of *cyp-14A5*. The authors use these experiments to suggest the potential specificity of the induction of CYP-14A5 by light. Given the established relationship between light and oxidative stress and the authors' later identification of ZIP-2, testing the effect of an oxidative stressor or pathogen exposure on transcription of *cyp-14A5* would further strengthen the validity of this statement and potentially shed some insight into the underlying mechanisms.

(2) The authors suggest that short-wavelength light more robustly increases transcription of *cyp-14A5* compared to equally intense longer wavelengths (Figure 2F and 2G). Here, however, the authors report intensities in lux of wavelengths tested. Measurements of and reporting the specific spectra of the incident lights and their corresponding irradiances (ideally, in some form of mW/mm² - see Ward et al., 2008, Edwards et al., 2008, Bhatla and Horvitz, 2015, De

Magalhaes Filho et al., 2018, Ghosh et al., 2021, among others, for examples) is critical for appropriate comparisons across wavelengths and facilitates cross-checking with previous studies of *C. elegans* light responses. On a related and more minor note, the authors place an ultraviolet shield in front of a visible light LED to test potential effects of ultraviolet light on transcription of *cyp-14A5*. A measurement of the spectrum of the visible light LED would help confirm if such an experiment was required. Regardless, the principal conclusions the authors made from these experiments will likely remain unchanged.

(3) The authors report an interesting observation that animals exposed to ambient light (~600 lux) exhibit significantly increased memory retention compared to those maintained in darkness (Figure 4). Furthermore, light deprivation within the first 2-4 hours after learning appears to eliminate the effect of light on memory retention. These processes depend on CYP-14A5, loss of which can be rescued by re-expression of *cyp-14A5* in mutant animals using a hypoderm-specific- and non-light-inducible- promoter. Taken together, the authors argue convincingly that hypodermal expression of *cyp-14A5* can contribute to the retention of the olfactory memory. More broadly, these experiments suggest that cell-non-autonomous signaling can enhance retention of olfactory memory. How retention of the olfactory memory is enhanced by light generally remains unclear. In addition, the authors' experiments in Figure 1B demonstrate - at least by use of the transcriptional reporter - that light-dependent induction of *cyp-14A5* transcription at 500 - 1000 lux is minimal and especially so at short duration exposures. Additional experiments, including verification of light-dependent changes in CYP-14A5 levels in the olfactory memory behavioral setup, would help further interpret these otherwise interesting results.

(4) The experiments in Figure 4 nicely validate the usage of the *cyp-14A5* promoter as a potential tool for light-dependent induction of gene expression. Despite the limitations of this tool, including those presented by the authors, it could prove useful for the community.

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